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Processing Capacity Under Perceptual and Cognitive Load: A Closer Look at Load Theory

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Variations in perceptual and cognitive demands (load) play a major role in determining the efficiency of selective attention. According to load theory (Lavie, Hirst, Fockert, & Viding, 2004) these factors (a) improve or hamper selectivity by altering the way resources (e.g., processing capacity) are allocated, and (b) tap resources rather than data limitations (Norman & Bobrow, 1975). Here we provide an extensive and rigorous set of tests of these assumptions. Predictions regarding changes in processing capacity are tested using the hazard function of the response time (RT) distribution (Townsend & Ashby, 1978; Wenger & Gibson, 2004). The assumption that load taps resource rather than data limitations is examined using measures of sensitivity and bias drawn from signal detection theory (Swets, 1964). All analyses were performed at two levels: the individual and the aggregate. Hypotheses regarding changes in processing capacity were confirmed at the level of the aggregate. Hypotheses regarding resource and data limitations were not completely supported at either level of analysis. And in all of the analyses, we observed substantial individual differences. In sum, the results suggest a need to expand the theoretical vocabulary of load theory, rather than a need to discard it.

Keywords: selective attention, processing capacity, perceptual load, cognitive load, hazard functions

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The ability to pay attention selectively is essential. Drivers have to focus on the road while ignoring irrelevant distractions from the sidewalk. Pedestrians must notice approaching cars while screening out noise from their headphones. Since its inception, modern research on attention has ascribed failures of selective attention to limitations on the observer's perceptual and cognitive resources, or capacity (Broadbent, 1958; Kahneman, 1973; Moray, 1967; Navon, 1977; Schweickert & Boggs, 1984; Shiffrin & Schneider, 1977; Taylor, Lindsay, & Forbes, 1967; Townsend & Ashby, 1978). Capacity plays a central role in a contemporary account of selective attention—load theory (Lavie et al., 2004; Lavie, 1995, 2000)—which offers an explanation for failures of selective attention. Load theory distinguishes two types of demands (load): perceptual and cognitive. Perceptual load is modulated by factors such as display size and cognitive load is modulated by factors such as demands on working memory.

Load theory assumes that perceptual and cognitive load have independent and opposite effects on the allocation of attentional resources and consequently on the efficiency of selective attention (Lavie, 2000; Lavie et al., 2004; Lavie, 1995; Lavie & Tsal, 1994; Lavie & de Fockert, 2003).

The present study addresses this assumption by using measures of performance that can speak to the issue of efficiency in a more direct and fine-grained manner than has typically been the case. Specifically, we employ measures of capacity based on the hazard function of the reaction time (RT) distribution (Townsend & Ashby, 1978, 1983; Townsend & Nozawa, 1995; Wenger & Gibson, 2004). We combine these with measures of perceptual sensitivity and response bias from signal detection theory (Swets, 1964; Macmillan & Creelman, 2005; Wickens, 2002), in order to simultaneously address load theory's assumptions regarding (respectively) real-time processing capacity and discriminability under variations in type and level of processing demands.

Perceptual Load and Processing Capacity

In a typical experiment in the study of perceptual load, participants are presented with a target letter that appears either alone (e.g., low perceptual load), or embedded among a number of non-target items (high perceptual load, e.g., Johnson, McGrath, & McNeil, 2002; Lavie, 1995; Lavie & Cox, 1997; Wilson, MacLeod, & Muroi, 2008). A distractor letter appears randomly in some nearby position, and the observer is instructed to respond to the target letter while ignoring the distractor. The critical regularity is that, under low perceptual load, observers respond slower in the incongruent relative to a neutral condition, while under high perceptual load this "flanker" interference (Eriksen & Eriksen, 1974) is virtually eliminated (see also Miller, 1991; Lavie, 1995; Lavie & Cox, 1997; Lavie, 2000). Comparable effects of perceptual load on selective attention have been observed with various other paradigms and manipulations of perceptual load. All of these ap-

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proaches converge on the same result—the elimination of distractor processing under high levels of perceptual load (for a review, see Lavie, 2005). According to load theory, high load drains attentional resources, thus leaving nothing left with which to process the distractor; in contrast, low perceptual load frees enough capacity that subsequently "spills over" to the distractor, resulting in the flanker interference. Also implied by load theory is the conjecture that perfect selective attention of the target is not possible when target processing does not utilize all of the available processing resources (Lavie, 1995). Therefore, perceptual load determines the failure or success of selective attention.

While the effect of perceptual load on selective attention is fairly robust (Dark, Johnston, Myles-Worsely, & Farah, 1985; Kahneman & Chajczyk, 1983; Miller, 1991), it is not without controversy (e.g., see Johnson et al., 2002; Paquet & Craig, 1997; Yantis & Johnston, 1990), and its logic of changes in processing capacity has been challenged several times (e.g., Eltiti, Wallace, & Fox, 2005; Hay, Milders, & Niedegeen, 2006; LaBerge, 1995, 2002; Miller, 1991; Paquet & Craig, 1997; Tsal & Benoni, 2010). For example, LaBerge (1995, 2002) proposed that increased load results in increased distractor inhibition and consequently leads to the elimination of interference. Miller (1991) demonstrated that it is possible to obtain the flanker effect under high perceptual load, suggesting that interference results not from the exhaustion of mental resources, but rather from an imbalance in processing-speed between target and distractor.

Another alternative account for perceptual load effects in terms of relative salience was proposed by Paquet and Craig (1997) and Eltiti et al. (2005). Paquet and Craig (1997) eliminated the flanker effect in low perceptual load displays by using category differences (e.g., digits as distractors and letters as targets in a flanker task). This manipulation presumably increased the target's salience and led to improved selectivity. In a similar vein, when Eltiti and colleagues (2005) made the distractor more salient than the target, reliable interference was obtained under high perceptual load. Eltiti et al. (2005) interpreted these results as suggesting that perceptual load hampers the quality of the input (e.g., a data limitation) rather than affecting the processing of the input (e.g., a resource limitation, see Norman & Bobrow, 1975).

These alternative accounts suggest that factors other than changes in processing capacity may determine the extent to which interference is obtained. Variations in perceptual load may not directly affect the allocation of attentional resources. Instead, they may simply affect the informational value of the stimulus. This possibility was noted by Lavie (Lavie & de Fockert, 2003; Lavie, 2005), who offered two pieces of evidence against such confound. First, in a study by Lavie and de Fockert (2003) the informational value of the stimuli was intentionally diminished by decreasing the target's physical size or contrast. Consequently the task became harder and RTs were prolonged. However, in spite of this presumed imbalance in relative speed-of processing between target and distractors (Miller, 1991), distractors still interfered with the target (Lavie & de Fockert, 2003). The results of this study are somewhat inconclusive because the data limitation manipulation was administrated only in conditions of low perceptual load. A second argument against such a confound is that increasing task difficulty by using an additional working-memory task (e.g., cognitive load) resulted in opposite effect to that of perceptual load: an increase in interference (Lavie et al., 2004; Lavie, 2000).

Cognitive Load and Processing Capacity

Load theory's conception of cognitive load is closely tied to demands on working memory. The central assumption is that placing demands on working memory results in less capacity to deal with distractors, hence larger interference is expected under high relative to low cognitive load conditions. In the paradigm developed to test effects of cognitive load on selective attention (e.g., Lavie et al., 2004), participants are presented first with a set of either one (low cognitive load) or six (e.g., high cognitive load) digits, which they are required to maintain in memory. Participants have to memorize the digits. A low perceptual load trial (e.g. a flanker task display) is then presented during the retention interval. Immediately after responding to the target, a probe digit appears, and observers indicate whether that digit was part of the memory set. The critical finding is that the level of interference in the flanker task is greater under high relative to low cognitive load (Lavie et al., 2004; Yi, Woodman, Widders, Marois, & Chun, 2004).

According to load theory, working memory functions to actively maintain task and action priorities. This assumption can be understood in terms of implications for processing capacity. As with perceptual load, effects of cognitive load may not stem from changes in processing capacity. An alternative account in terms of lateral inhibition has been proposed by LaBerge (1995, 2002). LaBerge suggested that an increase in cognitive load (e.g., as can be induced by increasing the difficulty of a secondary task) would be expected to result in reduced distractor inhibition due to the narrowing of attention to the target area. This explanation is based on an analogy to properties of neural circuits, and it assumes that lateral inhibition spreads outward from target sites to distractor sites and to sites of other tasks, and becomes weaker as the intensity of attentional activity at the target site decreases. Thus, it might be that the effects of cognitive load are not related to changes in the allocation of resources, but to the decrease in the observer's ability to inhibit their responses to distractors under high cognitive load.

To sum up, load theory specifies two mechanisms of selective attention. The first is a perceptual mechanism that is responsible for exclusion of irrelevant information in conditions of high perceptual load (Lavie, 1995; Lavie & Tsal, 1994). The second is a cognitive mechanism that governs attentional control in cases of low perceptual load, where irrelevant information is presumably available. Load theory assumes a complete independence between the two—a conjecture supported by the lack of interaction between perceptual and cognitive load effects (Lavie et al., 2004). The operation of both mechanisms relies on the assumptions that (a) each draws on a separate pool of resources (Navon, 1984); (b) these pools of resources are independent of one another; and (c) there is a fixed total level of conservation of resources, such that there will always be a trade-off between the resources allocated to the target and distractors. In addition, load theory assumes that in the case of perceptual load, allocation of resources is automatic. The observer has no control over the distribution of the remaining resources to processing the distractors.

Implications and Predictions

We used the assumptions of load theory to develop a set of six hypotheses regarding capacity, sensitivity, and response bias (see

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Table 1) under manipulations of perceptual or cognitive load.[1] Experiment 1 tests the hypotheses regarding the effects of perceptual load and Experiment 2 tests the hypotheses regarding the effects of cognitive load.

Before describing our hypotheses, a comment is in order regarding the measurement of selective attention. Three levels of target-distractor compatibility are typically used in studies of perceptual and cognitive load: congruent, incongruent, and neutral. This allows researchers to evaluate separately the contributions of interference (e.g., incongruent vs. neutral) and facilitation (e.g., neutral vs. congruent). Previous perceptual load studies (Lavie, 1995; Lavie & Cox, 1997; Paquet & Craig, 1997; Wilson et al., 2008) have used only interference as an index for selective attention. This was done due to problems in interpretability of the congruent condition (see e.g., Lavie, 1995; Estes, 1972). However, recent cognitive load studies have deployed the congruent condition and used congruency as a central measure (see Lavie et al., 2004, Experiments 1–3). The current experiments included all three compatibility levels. However, in framing our hypotheses, we used the separate conventional measures.

Hypotheses 1a and 1b concern effects at the level of mean RTs, predicting a requisite replication of earlier results (e.g., Lavie et al., 2004; Lavie, 1995, 2000). Obtaining these results is critical for the interpretability of the remaining analyses, in the sense that we need to establish that, when the data are examined using the analyses that are standard for experiments assessing perceptual and cognitive load, the standard set of results are obtained. Hypothesis 1a implies effect of congruency under low perceptual load, but not under high perceptual load with a reliable load $\times$ congruency interaction (Lavie, 1995). Hypothesis 1b entails a larger congruency effect under high cognitive load than under low cognitive load with a reliable load $\times$ congruency.

Hypothesis 2a provides a strong test of the central theoretical construct of the load theory: processing capacity. Load theory makes the assumption that changes in processing capacity determine the failures or successes of selective attention. This conjecture is based on the finding of a reliable difference in mean correct RT between incongruent and neutral conditions under low perceptual load, but not under high perceptual load. Note that this regularity does not necessarily provide any information on the relative effort observers invest in those two conditions. To address this issue directly, we use the hazard function on the RT distribution: a measure of capacity that has conceptual and statistical advantages relative to mean RT (see e.g., Townsend & Ashby, 1978; Townsend & Nozawa, 1995). First, it maps to the concept of capacity in a way that is much more direct than that allowed by mean RT. Second, it offers a finer-grained and more comprehensive view of performance than is possible with mean RT. The hazard function can be interpreted in the present context as the likelihood of the task being completed in the next instant conditional on it not yet having been completed (e.g., a formal definition appears in the results section). The cumulative version of this function possesses the desired characteristics for a measure of performance that quantify efficiency or work done in a unit of time. Hypothesis 2a thus takes the form of orderings on hazard functions for individual and aggregate observers in neutral and incongruent conditions.

The regularity of a reliable difference in mean RT (relative speed of completing the task) between incongruent and neutral conditions under low perceptual load suggests differences at the level of processing capacity. By this logic, changes in capacity due to congruency should be observable only under conditions of low perceptual load. Thus we would expect equality at the level of the hazard functions (i.e, the hazard function for neutral trials should not be statistically different from the hazard function for incongruent trials, $h_{H,N}(t) = h_{H,I}(t)$ under high perceptual load, and an inequality, namely ordering of the hazard functions [i.e., $h_{L,N}(t) > h_{L,I}(t)$] under low perceptual load. This pair of inequalities parallels the predictions for no congruency effect under high perceptual load and a reliable congruency effect under low perceptual load in Hypothesis 1a. The magnitude of the changes in the hazard functions should be ordered, such that: $\frac{hL,C(t)}{hL,I(t)} > \frac{hH,C(t)}{hH,I(t)}$. This ordering parallels the load $\times$ congruity interaction in Hypothesis 1a. In addition, given that the perceptual load manipulation is intended to tax capacity, we should obtain an ordering on the hazard functions across levels of load, holding congruency fixed, i.e., $h_{H,N}(t) < h_{L,N}(t)$ and $h_{H,I}(t) < h_{L,I}(t)$.

Hypothesis 2b concerns the effects of working memory load on selective attention. According to the load theory (Lavie et al., 2004; Lavie, 2000; Yi et al., 2004), larger mean RT differences between incongruent and congruent trials obtain under high cognitive load than under low cognitive load. This is due to better executive control or improved efficiency in dealing with distractors as levels of working memory load decrease. Thus, congruency effects on capacity should be observable, but less pronounced at low levels of cognitive load than at high levels of cognitive load. In terms of the hazard functions, this implies that $h_{H,C}(t) > h_{H,I}(t)$ and $h_{L,C}(t) > h_{L,I}(t)$. The magnitude of these differences should be ordered, such that $\frac{hH,C(t)}{hH,I(t)} > \frac{hL,C(t)}{hL,I(t)}$. The former pair of inequalities parallels the predictions for congruency effects under both low and high cognitive load; the latter reflects the interaction of load and congruency. In addition, given that cognitive load (like perceptual load) is intended to tax capacity, we again expect an ordering on the hazard functions across levels of load, holding congruency fixed, i.e., $h_{H,C}(t) < h_{L,C}(t)$ and $h_{H,I}(t) < h_{L,I}(t)$.

Hypothesis 3a addresses the assumption (e.g., Lavie & de Fockert, 2003; Lavie, 2005) that perceptual load places increased demands on capacity but does not affect the informational value of the stimulus. Essentially, the addition of letters should selectively affect the amount of capacity available for processing the target without affecting the amount of information from the stimulus, and thus its relative discriminability. Similarly, by this logic, target-distractor compatibility should also have no effect on the amount of stimulus information. Given this reasoning, and given that the paradigm used in a number of studies of perceptual load (e.g., Lavie, 2005, along with the current study) has been developed to maximize target-distractor discriminability, there should be no

[1] As with any verbal theory, there is an inherent ambiguity in the extent to which our interpretation of the theory is consistent with the veridicality of the theory, as intended by the framers of the theory. This being acknowledged, we believe that in the following hypotheses we captured the meaningful verbal statements conveyed by load theory in a way that allows a strong and fair test of the theory.

Table 1
Summary of the Load Theory Hypotheses Regarding the Effects of Congruency (e.g., Target-Distractor Compatibility) Under Perceptual and Cognitive Load on Mean RT, Hazard Functions, Sensitivity, and Response Bias

Expt.	Load	Hyp.	Type of measure/construct	High load	Low load	Interaction
1	Perceptual	1 (a)	RT (speed of processing)	–	+	+
1	Perceptual	2 (a)	hazard function (capacity)	–	+	+
1	Perceptual	3 (a)	d' (data limitations)	–	–	–
1	Perceptual	4 (a)	c (data limitations)	–	–	–
2	Cognitive	1 (b)	RT (speed of processing)	+	+	+
2	Cognitive	2 (b)	hazard function (capacity)	+	+	+
2	Cognitive	3 (b)	d' (data limitations)	–	–	–
2	Cognitive	4 (b)	c (data limitations)	–	–	–

Note. A plus sign implies the presence of congruency effect; a minus sign implies the lack of it. Expt. = Experiment number; Hyp. = Hypothesis number; Interaction = Congruency × Load.

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reliable differences in sensitivity (d') across levels of congruency and across levels of load. Moreover, the load × congruency interaction should not be reliable.

Hypothesis 3b addresses the assumption that cognitive load induces changes in processing capacity rather than affecting the salience of the target. Thus, the reasoning parallels that for Hypothesis 3a, under the assumption that variations in working memory demand selectively affect processing efficiency rather than stimulus discriminability. Hence, the predictions for sensitivity (d') are identical to those of Hypothesis 3a, with no reliable differences expected as a function of either load or compatibility, or their interaction.

Hypotheses 4a and 4b address the effects of perceptual and cognitive load (respectively) on decisional processes, as measured by the signal detection parameter c. As noted for Hypotheses 3a and 3b, variations in both types of load are assumed to selectively affect processing resources rather than the amount of information available from the stimulus. In addition, load theory (Lavie, 2005) assumes that processing resources are distributed among the display elements in an automatic fashion. Assume that this distribution strategy is invariant across levels of load. Then, along with the assumption that variations in load do not affect total stimulus information, the predictions are for equality of response criterion across levels of load and across levels of stimulus compatibility with no interaction between load and congruency.

Thus, the two sets of hypotheses allow us to frame simultaneous predictions for performance at four levels of analysis: mean RT, capacity (hazard function of the RT distribution), response sensitivity (d'), and response bias (c). Critical here is the idea that the test of load theory rests on the results of the entire set of dependent variables, rather than just one. Thus, we provide the potential for what we believe is one of the strongest and most comprehensive tests to date of the assumptions of load theory.

There are precedents for using SDT in the study of attention (e.g., Bashinski & Bacharach, 1980; Estes, 1982; Shaw & Shaw, 1977). Recently, Macdonald and Lavie (2008) used SDT measures to study the effects of perceptual load in a modified version of the inattentional blindness paradigm. Consequently, it is worth considering how our application differs, as we believe it does, in important ways. First, Macdonald and Lavie used a dual-task to test the influence of perceptual load on the detection of a peripheral object. In contrast, as we sought to address specific implications of the assumptions of load theory, we used the flanker task and experimental manipulations that have been widely used in studies of perceptual load (Lavie, 1995). Thus, we are able to obtain SDT and RT measures within the same task that has been used in the development of load theory. Second, our approach allows us to test for effects of congruency, a critical source of evidence in the development of load theory. Third, we believe that the assumptions of load theory actually lead to opposite effects with respect to changes in d' than has been predicted by Macdonald and Lavie (2008). Our rationale (presented explicitly in our hypotheses section) is based on the distinction between resource and data limitations which is fundamental to load theory (Lavie & de Fockert, 2003; Lavie, 2005). Finally, we tested the hypotheses both at the aggregate and at the individual level. Sufficient data were obtained to allow analyzes of all measures to be performed at the level of individual observer.

Experiment 1

Experiment 1 tested the assumptions of load theory regarding the effects of perceptual load [Hypotheses 1a–4a], using both the standard measures of performance (mean RT) and measures that are more tightly bound to the concept of capacity than is mean RT (see relevant discussions in Townsend & Ashby, 1978; Townsend & Nozawa, 1995; Wenger & Gibson, 2004). Consequently, the experiment was designed as a partial replication and extension of those reported by Lavie et al., 2004 and Lavie, 1995. In addition, we estimated SDT measures (Macmillan & Creelman, 2005; Swets, 1964; Wickens, 2002), to test for the effects of perceptual load on sensitivity and response bias [Hypotheses 3a and 4a].

Methods

Participants

Eight participants were recruited from the student population of The Pennsylvania State University. All reported normal or corrected-to-normal vision, normal hearing, and unimpaired use of both hands. Participants were paid at the rate of $8/hr.

Apparatus and Stimuli

A PC-compatible computer with a 19 in (diagonal) color CRT set to a resolution of 1024 × 768 pixels was used to present all

stimuli and record all responses. Stimulus onsets were synchronized with the vertical retrace of the monitor (refresh rate of 100 Hz), and responses were timed to ± 1 ms using Macromedia's Authorware software (Macromedia, 2007). Observers viewed the display at a fixed distance of 76 cm through a black viewing tube, used to restrict the light available to observers' eyes to only the light emitted by the display. One end of the viewing tube was placed against the display while the other was placed against the observer's chin rest.

Stimuli (targets and distractors) were alphanumeric characters, presented using New Times Roman (font). All letters were presented in a light gray color on a black background. Target letters (lower case x or z) subtended a visual angle of 0.48° horizontally and 0.67° vertically. Distractor letters (upper case X, Z, or P) subtended a visual angle of 0.57° horizontally and 1.05° vertically. Non-target letters (lower case k, s, m, v, and n) each subtended visual angles of 0.48° horizontally and 0.67° vertically.

Design

Experiment 1 was designed as 2 (perceptual load: high, low) × 3 (congruency: congruent, neutral, incongruent) full factorial within-participants design. The dependent variables were mean RT, hazard function of the RT distributions, and the SDT parameters for sensitivity (d') and response bias (c).

Procedure

Each 1-hr session began with a 5-min period of dark-adaptation. All trials were initiated by the computer and began with a light-gray fixation dot that appeared at the center of the screen for 1 s. This was replaced by the target display, which was presented for 50 ms. The two possible target alternatives were presented with equal frequency in each block of trials. In the low load trials, the target was presented alone in any of six possible locations along a central horizontal row subtending 4.9°, with each possible location used with equal frequency. In the high load trials, the target was present along with five non-targets. In a congruent trial, the target and the distractor were assigned the same response. In an incongruent trial, the target and distractor required different responses. In a neutral trial, the distractor was the letter P, which had no response-assignment. On all trials, observers were instructed to respond to the target while ignoring the distractors.

Participants responded using the number keys at the top of the standard PC keyboard, pressing the "0" key with their middle finger and the "9" key with their index finger. No response feedback was provided. Conditions of high and low perceptual load were presented in separate and alternating blocks. Half of the participants started with a high load block, while the other half started with a low load block. Each participant completed three days of testing, with each session composed of ten blocks of 72 trials.

Results

Each section that follows corresponds to one of the hypotheses in Table 1. Within each section, we first report results at the level of the aggregate (averaged across participants), and then consider the results at the level of individual observers. An α-level of 0.05 was adopted for all analyzes.

Mean RT

Prior to analysis, RTs shorter than 150 ms or longer than 2000 ms were excluded (a total of 4% of all trials across all observers). We followed Lavie and Cox (1997) and Wilson et al. (2008) in omitting from analysis the congruent distractors because of the difficulty of separating response-based facilitation effects from feature-based priming (see also, Estes, 1972). We used the conventional interference measure, which is the cost in performance produced by the incongruent condition relative to the neutral condition (Lavie, 1995). Mean RTs for correct responses were estimated for each observer at each level of load and congruency, and these data were analyzed using a 2 (load: low, high) × 2 (congruency: incongruent, neutral) repeated-measures analysis of variance (ANOVA), first collapsing across observers, then separately for each of the observers. The results of these analyzes are presented in the upper portion of Table 2.

At the level of the aggregate, the main effects of both load and congruency were reliable, as was the interaction (see upper left of Table 2). Consistent with the predictions of Hypothesis 1a, a reliable interaction between congruency and load was obtained. The interference from the incompatible distractor was reliable under low (32 ms) but not under high perceptual load (9 ms). These outcomes replicate the standard patterns for the manipulation of perceptual load (e.g., Lavie, 1995).

The results reported so far are averages across observers, which may or may not be reflective of regularities at the level of individual observers (Estes, 1956; Maddox, 1999). Consequently, the preceding analysis was repeated at the level of the individual observers. All eight observers were affected by the manipulation of load (i.e., slower RTs under high perceptual load). Critically, however, only four out of the eight observers' performance reflected the load × congruity interaction pattern—this being the pattern expected according to load theory [Hypothesis 1a].

The individual variations can be seen in the distribution of points in Figure 1. Here were plot, for each observer, the mean difference in RTs between the neutral and incongruent conditions at the two levels of perceptual load. Had all observers conformed to the pattern predicted by load theory, [Hypothesis 1a], their data points should all have been below the diagonal and very close to the zero line. However, this is not the case. Some (Observers 1 and 8) showed reliable interference in both the low and high perceptual load conditions. Others (Observer 3) exhibited the expected ordering of mean RTs, but at levels that did not reach criterion for statistical reliability. We interpret this level of variability in terms of individual differences (Estes,1956; Siegler, 1987; Maddox, 1999; Townsend & Fific, 2004), suggesting the strong possibility that there are critical variations in the ways in which attention can be allocated, and that can only be revealed by examining data at the level of individual observers. This being acknowledged, it is worth emphasizing that, when we average our data across observers, the critical results in earlier work (e.g., Lavie, 1995) are replicated: thus, the predictions of Hypothesis 1a were supported at the group level, although we found some violations of it in the data of individual observers.

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Table 2
Experiment 1: ANOVA on Mean RT, for the Aggregate and Separately for Each of the Observers

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
A	L	1	556.71	261.89‡					
	C	1	182.92	23.64†					
	L × C	2	80.42	23.90†					
1	L	1	10792054	292.49‡	2	L	1	16837304	324.71‡
	C	1	469235	12.72‡		C	1	16524	0.32
	L × C	2	895	0.02		L × C	2	223744	4.31*
3	L	1	5259694	52.28‡	4	L	1	8330161	694.46‡
	C	1	1676263	1.66		C	1	72314	6.03†
	L × C	2	31761	0.32		L × C	2	44674	4.72*
5	L	1	13492050	296.62*	6	L	1	5660313	45.80‡
	C	1	199652	4.39*		C	1	187859	1.52
	L × C	2	265288	5.83*		L × C	2	361929	2.93
7	L	1	8658436	739.91‡	8	L	1	7526813	316.91‡
	C	1	369608	31.24‡		C	1	271681	11.44‡
	L × C	2	68895	7.35‡		L × C	2	33460	1.41

Note. L = load; C = congruency; A = aggregate; O = observers.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

Capacity

Next we consider the predictions of Hypothesis 2a regarding changes in processing capacity. To do this, we take advantage of the formal characterization of capacity, as a general characteristic of human information processing, originally developed by Townsend and Ashby (1978). Specifically, in that work, and the work that has followed (Townsend & Ashby, 1983; Townsend & Nozawa, 1995; Wenger & Townsend, 2000; Wenger & Gibson, 2004; Wenger & Townsend, 2006), overall system capacity is characterized in terms of the instantaneous intensity with which the system under study can process inputs that vary in terms of the level of demands placed on that system. This is done at the level of the hazard function of the RT distribution—the conditional probability of completing processing in the next instant of time, given that processing has not yet completed:

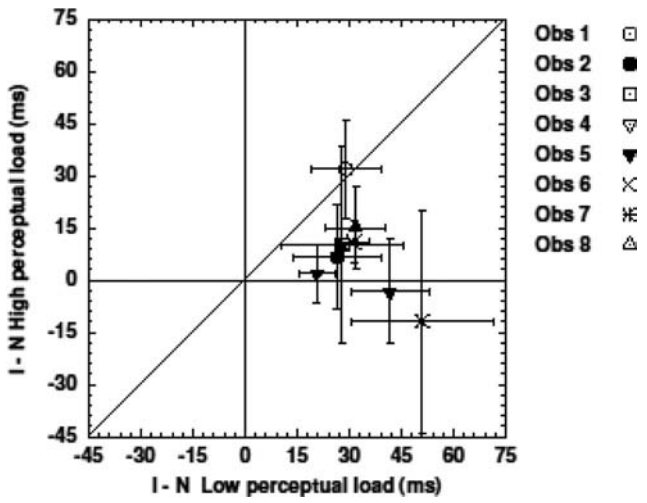


Figure 1. Experiment 1: Differences in mean RTs between incongruent (I) and neutral (N) conditions, under low and high perceptual load, for each of the eight observers. Error bars represent the standard error of the mean.

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \le T \le t + \Delta t | T \ge t)}{\Delta t} = \frac{f(t)}{S(t)}$$

Statistical hypothesis testing at this level of analysis is accomplished by way of semi-parametric regression models (Cox, 1972; Grambsch & Therneau, 1994; Therneau & Grambsch, 2000) known as proportional hazards models (see also Allison, 1995; Collett, 1994).This approach has multiple strengths, including the ability to assess reliable orderings at the level of the hazard function due to individual and interacting effects of independent variables, and variations in those orderings as a function of point in the RT distribution.

For the sake of consistency with the mean RT analysis, the capacity analysis was restricted to neutral and incongruent conditions under high and low perceptual load. We applied the Cox model (Cox, 1972) to the RT distributions in those conditions. In order to analyze the aggregate data, we first examined the individual observer data to determine if there were any reliable interactions of RT with time (e.g., inconsistencies in the orderings on RT hazard functions in different time values). Data for two of the observers (Observers 3 and 6) showed such interactions. Consequently, we restricted the aggregate analysis to the data from the remaining six observers, using stratification to account for heterogeneity due to individual differences (Allison, 1995). The results are presented in the top portion of Table 3.

In a similar manner to the ANOVA performed on mean RT (at the aggregate level), the results of this analysis revealed a reliable load × congruity interaction.[2] In addition, the effects of both load and congruency were reliable, with increases in load leading to relative reductions in capacity and congruency leading to relative increases in capacity. To better understand the two-way interac-

[2] Technically, the reliable interaction at the level of the hazard function implies a reliable interaction at the level of the mean (Townsend, 1990). However, the specifics of the methods of estimating these effects at these different levels do not guarantee that this result will always hold in experimental data.

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Table 3
Experiment 1: Proportional Hazards Analyzes for the Aggregate and Individual Observers

O	Effect	β	*SE*	χ^2	O	Effect	β	*SE*	χ^2
A	L	−1.83	0.06	533.64‡					
	C	0.64	0.07	58.32‡					
	L × C	−0.55	0.09	25.54†					
1	L	−0.69	0.06	113.38‡	2	L	−0.63	0.06	105.79‡
	C	0.23	0.06	13.93‡		C	0.13	0.06	4.95*
	L × C	−0.11	0.09	1.58		L × C	−0.12	0.09	2.01
3	L	−0.28	0.11	6.13*	4	L	−0.10	0.07	237.56‡
	C	0.34	0.09	13.79‡		C	0.28	0.06	20.58
	L × C	−0.55	0.16	11.46†		L × C	−0.28	0.89	9.79†
5	L	−0.62	0.06	97.86‡	6	L	−0.23	0.09	6.49*
	C	0.28	0.06	21.71‡		C	0.16	0.09	3.49
	L × C	−0.30	0.09	11.71‡		L × C	−0.23	0.13	3.02
7	L	−0.75	0.05	222.01‡	8	L	−0.70	0.07	102.34‡
	C	0.44	0.05	77.97‡		C	0.26	0.06	17.05‡
	L × C	−0.36	0.07	26.31‡		L × C	−0.19	0.09	4.41*

Note. L = load; C = congruency; A = aggregate; O = observers.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

tion, we fit the model to the data separately for each load and congruency condition, and the results of the aggregate analysis are presented in the bottom portion of Table 4.

The low perceptual load condition showed a reliable change in processing capacity, whereas the high perceptual load condition exhibited a smaller change. Confidence intervals derived from the standard errors of the βs revealed that the difference in the βs was reliable. In addition, we tested the hazard ratio between low and high perceptual load for neutral and incongruent conditions separately. These ratios were both significant: for neutral [$\beta = -0.656$, $\chi^2(1, 4) = 690.52$, $p < .0001$], and for incongruent conditions [$\beta = -0.809$, $\chi^2(1, 4) = 1079$, $p < .0001$].

When considered in terms of capacity (i.e., at the level of the hazard functions), the aggregate data were generally consistent with the predictions of load theory (e.g., Lavie, 1995). Specifically, the effect of congruency was greater at low levels of load than it was at high levels of load. Note, however, that according to load theory, there should be no effect of congruency with this set of stimuli under high perceptual load (Lavie, 1995). Instead, we obtained a small but reliable increase.

We next consider the capacity analyses at the level of the individual observers. The lower portion of Table 3 presents the results of this analysis. Recall that the capacity analysis (and all the analyses pertinent to the perceptual load) are held with neutral

Table 4
Experiment 1: Proportional Hazards Analyzes for the Effect of Congruency, Separately for Each Level of Load, for the Aggregate and the Individual Observers

Observer	Load	β	*SE*	χ^2	% Change
Aggregate	Low	0.200	0.07	69.95‡	22
	High	−0.049	0.060	3.95*	5
1	Low	0.192	0.061	9.76†	21
	High	0.013	0.065	4.10*	1
2	Low	0.127	0.060	4.45*	15
	High	0.008	0.06	0.02	0
3	Low	0.284	0.088	10.39†	33
	High	−0.297	0.13	5.17*	−25
4	Low	0.244	0.06	16.18†	27
	High	−0.006	0.064	0.01	0
5	Low	0.23	0.0619	13.85†	25
	High	−0.01	0.063	0.09	0
6	Low	0.158	0.071	4.93*	17
	High	−0.065	0.081	0.65	−6
7	Low	0.35	0.05	48.98†	42
	High	0.093	0.049	3.5	9
8	Low	0.246	0.063	15.18†	27
	High	0.0719	0.067	1.12	7

Note. % change = $(e^{\beta} - 1) \times 100$.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

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and incongruent conditions. Five of the eight observers exhibited a reliable load × congruency interaction, either across the entire range of the distribution or (for Observers 3 and 6) within the limited range, over which there were no reliable interactions of the experimental factors with time. Consequently, at the level of the hazard function we note the presence of individual differences observed first in the mean RT data. These individual differences are not due to low statistical power as we collected sufficient amount of data from each participant to guarantee high power.

We also estimated the effect of interference (e.g., the hazard ratio for neutral and congruent trials) in low and high perceptual load conditions, separately for each observer. The estimates for the proportionality due to interference are plotted in Figure 2. The estimations for Observers 3 and 6 excluded regions of the distributions that showed time covariates. Two points are critical here. The first is that any benefits due to interference are greater under low perceptual load than under high perceptual load. The second is that, in all cases, the sign of the estimate under low load was negative, while for the majority of cases the sign of the estimate under high load was positive. This indicates that load produced benefits in processing under high load while extracting costs under low load, outcomes that are generally consistent with load theory.

Sensitivity and Response Bias

We next consider the data with respect to Hypotheses 3a and 4a. Recall that, following the assumptions of load theory, perceptual load is not predicted to affect either sensitivity or bias, at either low or high loads. For the purposes of these analyzes, a "hit" was defined as responding "x" to the target "x" and a "false Alarm" was defined as responding "x" to the target "z" (see also, Yi et al., 2004). Confidence intervals for the SDT parameters were derived by using Gourevitch and Galanter method (1967).

The pertinent aggregate results for Hypothesis 3a can be seen in Table 5. At the level of the aggregate, better target discriminability was found under low relative to high perceptual load. This result is rather surprising and runs counter to the predictions of Hypothesis 3a (no reliable differences in sensitivity). The immediate implication is that perceptual load appears to affect the salience of the target. Moreover, better discriminability was observed under neutral than under incongruent conditions across both levels of perceptual load. Figure 3a plots the difference in d' [e.g., neutral minus incongruent with error bars derived from the Gourevitch and Galanter (1967) statistical test] in low perceptual load against the same measure in high perceptual load, for individual observers. As can be seen, while all but one of the differences (observer 7, high load) were reliably different from zero, there were substantial individual differences as a function of load. For four observers (1, 5, 6, 7), sensitivity was higher under low relative to high perceptual load. The opposite (superior sensitivity under high perceptual load) was observed for two observers (3 and 8). And for two observers (2 and 4), sensitivity differences were equal across the two levels of load. Based on these results we submit that the predictions of Hypothesis 3a were refuted. Specifically, it appears that, contrary to the assumptions of load theory, perceptual load does in fact induce data limitations.

Table 5
Experiment 1: Results of the Aggregate Analysis of the Sensitivity and Bias Measures

Measure	Effect	*df*	*MSE*	*F*
d'	L	1,7	7.19	0.14*
	C	1,7	7.51	0.07†
	L × C	1,7	1.59	0.02
C	L	1,7	0.03	0.008
	C	1,7	1.59	0.011
	L × C	1,7	1.10	0.008

Note. d' = sensitivity; C = bias; L = load; C = congruency.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

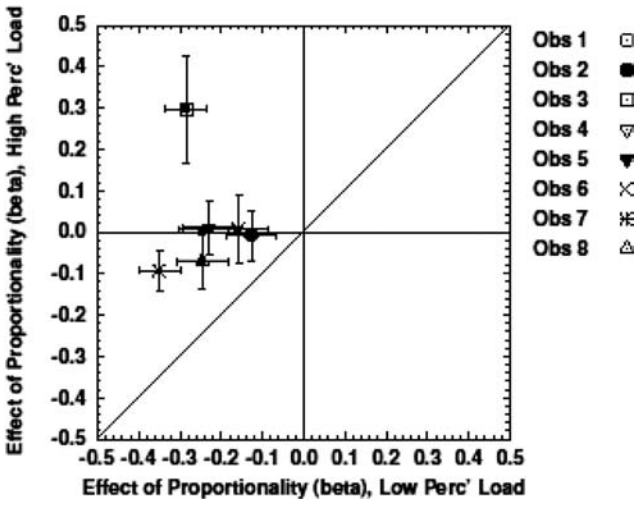


Figure 2. Experiment 1: Estimated effects of congruency under high and low perceptual load, for each of the observers. Note: error bars indicate standard deviation of the parameter estimation.

The analysis of the aggregate response bias data pertinent to Hypothesis 4a, (see Table 5) revealed no reliable effects of perceptual load or congruency. However, the individual analysis gives a different picture. Figure 3b plots the differences in response bias (e.g., neutral–congruent) for low perceptual load condition against the same measure for high perceptual load condition. Four distinctive patterns are discernible. Two observers (3, 7) showed no changes in response bias whatsoever. Three observers (1, 2, 4) exhibited a positive increase in response bias under both low and high perceptual load. Two observers (6, 8) revealed increases in response bias under low perceptual load but decreases under high perceptual load. One observer (5) showed decreases under both levels of perceptual load. In sum, although the results at the level of the aggregate are consistent with the predictions of Hypothesis 4a, the results at the level of the individual observers are not.

Discussion

Experiment 1 provided a test of four hypotheses pertinent to the effect of perceptual load, derived from load theory (Lavie, 1995). The first (at the level of mean RT) was the standard pattern for the effect of perceptual load. The remaining three were novel, and addressed capacity, sensitivity, and response bias. Table 6 sum-

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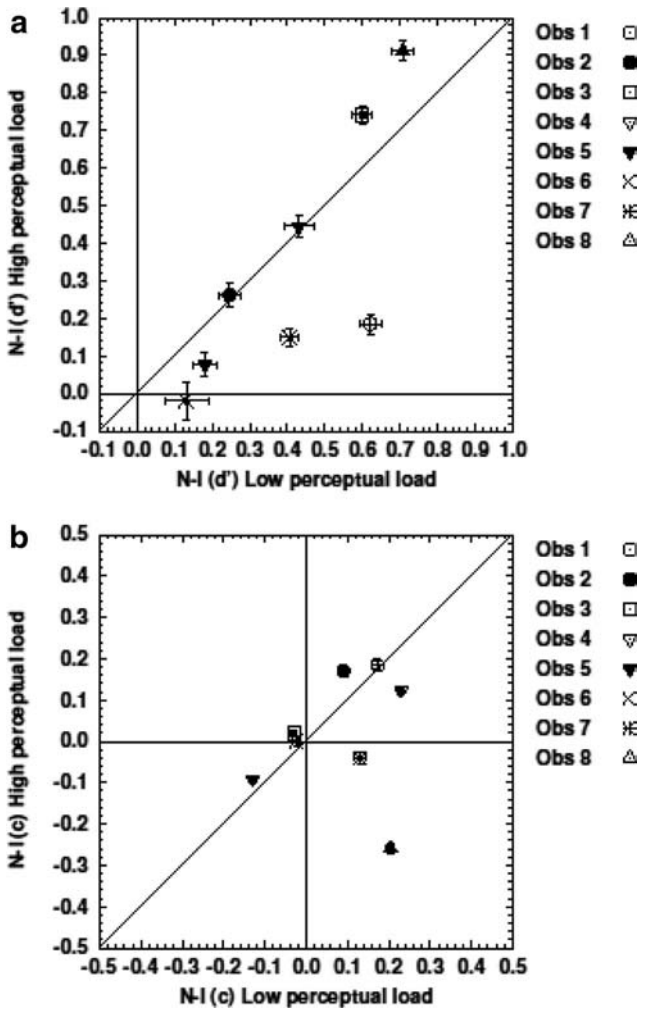


Figure 3. Experiment 1: differences in sensitivity (d'; Panel a) and in response bias (Panel b) between neutral and incongruent conditions for individual observers under low and high levels of perceptual load. Note: error bars indicate 95% confidence intervals (see for, Gourevitch and Galanter, 1967).

marizes the results for all four hypotheses, at the level of the aggregate and the level of the individual observers.

Hypothesis 1a captured the requisite conditions for making all of the results of the present effort both informative and relevant. That is, the hypothesis stated that, at the level of the aggregate, the regularities both predicted by load theory and generally obtained in studies of those predictions would hold. Specifically, the critical interaction of load and congruency—relatively improved selective attention under high versus low perceptual load—was confirmed at the level of the mean RT. However, we found considerable variability at the level of the individual observers. The characteristic signature of load theory—a reliable load × congruity interaction—was recorded for only half of the observers. This is quite remarkable given the robustness of the effect at the level of the group. Why does the aggregate analysis differ from the individual results? This is most likely not due to lack of statistical power, as we collected a sufficient amount of data to guarantee high statistical power. Each of our observers participated in more than 3,400 experimental trials.

We submit that the aggregate results are only representative of a general trend. This trend does not always reflect the psychological phenomena at the individual level. A priori it is reasonable to assume that observers vary with respect to the initial amount of processing capacity they have. The present results attested to that assumption. Given this natural variability in processing capacity, different patterns of selective attention are possible. Assume an observer with a large amount of processing capacity at his or her disposal. According to the logic of load theory itself, this participant should suffer large failures of selective attention not only under low perceptual load but also under high perceptual load. In contrast, an observer who is very poor in processing capacity should reveal perfect selective attention under both low and high perceptual load. An additional source of complexity unfolded in our data, is that capacity is not always a "linear" variable, and for some observers might radically change with response speed. Overall, the predictions of the load theory aim at describing an average observer. However, each observer has their own level of initial processing capacity. Thus, when looking at individual observers, different patterns of attention allocation are both logically and empirically possible. Generally, individual differences in the field of attention are given limited consideration or are completely overlooked (although, see Townsend & Fific, 2004, for an important exception). The present results highlight their existence and potential importance. There are examples in other fields of cognitive psychology demonstrating how the averaged data did not fit to the individual data (Estes, 1956;Maddox, 1999).

Hypothesis 2a directly addressed the concept of processing capacity that is fundamental to the load theory, by analyzing the data at the level of the hazard function of the RT distribution. At the aggregate level, most of the predictions of Hypothesis 2a were supported.[3] As expected, the effect of perceptual load on performance was mirrored in the reliable orderings on the hazard functions i.e., $h_{H,N}(t) < h_{L,N}(t)$ and $h_{H,I}(t) < h_{L,I}(t)$. Placing perceptual demands on performance required more processing capacity.

The hazard functions under low perceptual load were ordered in the predicted direction [i.e., $h_{L,N}(t) > h_{L,I}(t)$], reflecting a dramatic change in processing capacity (e.g., 22%) due to congruency. This effect corresponded to the notable failure of selective attention under low perceptual load, typically gouged by differences in mean RTs (Lavie, 1995; Miller, 1991). We also found a small but reliable change (e.g., 5%) in processing capacity under high perceptual load. This was reflected in a reliable ordering on the hazard functions [i.e., $h_{H,N}(t) > h_{H,I}(t)$], signaling a small but reliable failure of selective attention under high perceptual load. Although Hypothesis 2a did not predict such a change on the basis of previous results (Lavie, 1995), this outcome is still consistent with load theory. In particular, magnitude of changes in processing

[3] It is important to note however, that the analysis was confined to those observers who did not exhibit time-dependent covariates: an interactions of load with time. Thus, the generality of the conclusions drawn is limited to 6 out of 8 of the observers. Note that similar results observed when the data of the entire subset of observers were analyzed and time covariates were estimated.

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Table 6
Experiment 1: Summary of the Outcomes With Respect to Each of the Four Hypotheses

	Hypothesis			
Observer	1(a)	2(a)	3(a)	4(a)
Aggregate	∨	∨	*	∨
1	*	*	*	*
2	∨	*	*	*
3	*	∨	*	∨
4	∨	∨	*	*
5	∨	∨	*	*
6	*	*	∨	*
7	∨	∨	*	∨
8	*	∨	*	*

∨ prediction holds. * prediction fails.

capacity was ordered such that $\frac{hH,C(t)}{hH,I(t)} > \frac{hL,C(t)}{hL,I(t)}$. This asymmetry in processing capacity parallels that of the load × congruency interaction found in mean RTs and thus lends support to the central assumption of load theory.

Tested at the individual level, the support for Hypothesis 2a appeared to be less robust, with large differences across individual observers. With respect to the effects of congruency, these ranged from 42% to 15% under low perceptual load, and from 0% to 9% under high perceptual load. A critical point, however, is that the reliable difference for the high load condition was observed for only two of the observers, suggesting that the majority of the observers showed the pattern predicted by load theory.

Hypothesis 3a predicted that perceptual load does not diminish the informational value of the stimulus (Lavie & de Fockert, 2003; Lavie, 2005). To address this prediction, we estimated d' values under all experimental conditions. If perceptual load is affecting resources but not data, then sensitivity should have remained invariant across levels of load and congruency. In contrast, we found, at both the level of the aggregate and the level of the individual observers, that both load and congruency affected observers' sensitivity. Decreasing d' values as a function of load and congruency are strong evidence for data limitations (Norman & Bobrow, 1975). However, the interaction of load × congruency was not reliable.

Hypothesis 4a predicted that observers do not change their response bias when dealing with perceptual load. Thus, c was predicted to remain invariant across levels of load and congruency. Consistent with this hypothesis we did not find any reliable changes in c due to load congruency or their interaction at the level of the aggregate. However, at the level of the individual observers, we found substantial variability with some observers showing small but reliable differences in c due to interference.

Considered on the whole, either at the level of the aggregate or at the level of the individual observers, the data from Experiment 1 provide at best limited-to-medium support for the predictions of load theory (see Table 6). This overall limited level of support suggests serious questions with respect to the overall validity of load theory, in spite of the fact that, for one of the dependent measures (mean RT), at one level of analysis (the aggregate) the data from Experiment 1 represent a replication of standard results in studies of the effects of perceptual load and are thus completely consistent with load theory.

Experiment 2

Experiment 2 parallels the structure of Experiment 1 and provides a test of the predictions of load theory with respect to the effects of cognitive load. Specifically, Experiment 2 tested the predictions of Hypotheses 1b–4b, and was modeled on the experimental methods used in Lavie et al. (2004).

Methods

Participants

Eight participants, none of whom participated in Experiment 1, were recruited from the student population at the Pennsylvania State University. They were paid at the rate of $8/hr. All reported normal or corrected-to-normal vision, normal hearing, and unencumbered use of both hands.

Apparatus and Stimuli

The apparatus was identical to that used in Experiment 1. Stimuli (target, distractors, and the memory set) were alphanumeric characters, presented using Times New Roman font. Six asterisks were used to mask the digits. The target and distractors letters were presented in a light gray color on a black background.

All combinations of target identities, target positions, distractors identities, and distractor positions were counterbalanced. The digits comprising the cognitive load were randomly chosen from the 1–9 set. Each digit was equally likely to appear in the memory-set of each trial in random order under the constraint that the same digit could not appear in sequential order. A single digit served as a memory probe. On half of the trials, this probe was from the memory set, and on the other half of the trials it was not. For those trials on which the probe was from the memory set, the location of the item in the set was determined randomly and with equal likelihood across the six possible positions.

Design

The experiment was designed as 2 (cognitive load: high, low) × 3 (congruency: congruent, neutral, incongruent) full factorial within-participant design. The dependent variables were mean RTs, the hazard function of the RT distribution, and the SDT parameters for sensitivity (d') and response bias (c).

Procedure

Each 1-hr session began with a 5-min period of dark-adaptation. Each trial began with a light-gray fixation dot at the center of the screen for 500 ms. This was replaced by the memory set, presented for either 500 ms (low cognitive load) or 2 s (high cognitive load). The memory task required observers to memorize the six digits. Each digit subtended a visual angle of 0.57° horizontally and 1.05° vertically. The digits were presented in a horizontal row at the center of the display. In low load conditions, one digit was presented at the center of the display. Six asterisks were then used to mask the digits presented for 750 ms (low cognitive load) or 2.5 s

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(high cognitive load). A fixation point presented for 500 ms then appeared. It was preceded by a 100 ms presentation of the selective attention task display. The target letter subtended a visual angle of 0.48° horizontally and 0.67° vertically and was equally likely to be x or z. It was presented in a lowercase with equal probability in any of six possible locations along a central horizontal row subtending 4.9°. A distractor letter (x, z, or p) subtending a visual angle of 0.57° horizontally and 1.05° vertically appeared as a capital letter either above or below the row. In each trial, a target (e.g. x or z) and a distractor (e.g., X, Z, or P) were presented. Observers responded by using their right hand to press "0" on the numeric keypad if the target letter was "z", or "9" if the target was "x".

Responses to the selective attention task were recorded within 2 s. Then, a memory probe was presented on the screen for 3 s or until the observer responded. Observers used their left hand to give the responses for the memory probe. They pressed "q" if the probe digit was present or "w" if the probe digit was absent from the memory set. Conditions of high and low cognitive load were blocked and presented in an alternate fashion. Half of the participants started with a high load block, whereas the other half first started with a low load block. Each participant completed four days of testing. Each session was composed of seven blocks of 72 trials each. In total each participant performed in 28 blocks of trials, for a total of 2016 trials. The first block of the first day was considered as practice and thus discarded from analysis. A short break of 1 min separated between each block. The order of trials within block was determined randomly.

Results

As was the case for the presentation of the results for Experiment 1, each section that follows corresponds to one of the hypotheses in Table 1. Within each section, we first report results at the level of the aggregate (averaged across participants), and then consider the results at the level of individual observers.

Mean RT

Prior to analysis, RTs shorter than 150 ms or longer than 2000 ms for both the memory task and the selective attention task were excluded (a total of 3% of all trials across all observers). Before proceeding with the complete set of hypothesis tests, it was first necessary to confirm that the working-memory load manipulation was successful. Indeed, in the memory task, all observers showed better performance when retaining one rather than six digits, (all ts > 4.53 and all ps < 0.05). Next, we tested for the effect of cognitive load on the selective attention task (Lavie et al., 2004). Only trials where the observer was correct both on the memory task and on the selective attention task were included. Following Lavie et al (2004) we used only congruent and incongruent conditions.[4]

Mean RTs for correct responses were estimated for each observer at each level of load and congruency, and these data were analyzed using a 2 (load: low, high) $\times$ 2 (congruency: incongruent, congruent) repeated-measures ANOVA, first collapsing across observers, then separately for each of the observers. At the level of the aggregate, the main effect of congruency and its interaction with load were reliable (see upper left of Table 7). Consistent with the predictions of Hypothesis 1b, a reliable interaction between congruency and load was observed. The congruency effect under high cognitive load (176 ms) was larger than that under low cognitive load (121 ms). These results replicate the pattern reported by Lavie et al. (2004) and Yi et al. (2004) and are thus consistent with the predictions of Hypothesis 1b.

The bottom part of Table 7 presents the individual results. Noticeably, five out of eight observers showed a reliable load $\times$ congruency interaction—the pattern expected according to load theory [Hypothesis 1b]. Figure 4 illustrates the individual variations. It plots the differences in mean RT between congruent and incongruent conditions for the high cognitive load condition against the same measure for the low cognitive load condition. The diagonal represents equal levels of failures of selective attention under low and high cognitive load. Two observers (2, 8) were very close to both the zero point and the diagonal, showing no effects of cognitive load on selective attention. Five observers (1, 4, 5, 6, 7) were reliably above the diagonal and far off the zero point, and thus exhibiting the pattern of attention predicted by load theory (Lavie et al., 2004). One observer (3) conformed to the general trend but did not show a reliable load $\times$ congruency interaction. Thus, although the aggregate results lend support to Hypothesis 1b, the form and magnitude of the individual differences raise doubts as to the overall strength of that support.

Capacity

Next we turn to the predictions of Hypothesis 2b regarding changes in efficiency due to cognitive load. We restricted our capacity analysis to congruent and incongruent conditions under low and high cognitive load. In order to analyze the aggregate data, we first examined the individual observers' data to determine if (as was the case for two observers in Experiment 1) there was any evidence for time-dependent covariates. None of the participants have evidenced this interaction. Thus, we included all of the data for all of the participants in the analysis.

Consistent with the analyses of the mean RTs, the results of this analysis revealed a reliable congruency $\times$ load interaction at the aggregate level. These results can be seen at the top part of Table 8. Moreover, the effects of both load and congruency were reliable, with load leading to a relative decrease in capacity and congruency leading to a relative increase in capacity. To better understand the two-way interaction, we fit the model to the data separately for each load and congruency condition, and the results of the aggregate analysis are presented in the bottom portion of Table 9.

Reliable differences in capacity were observed for both the low and high cognitive load conditions, with the proportional difference being largest in the high cognitive load condition. Thus, the results at the level of the aggregate are partially consistent with the predictions of the load theory (Lavie et al., 2004). In addition, we tested the hazard ratio between low and high cognitive load for congruent and incongruent conditions separately. The ratio for congruent condition was reliable [$\beta = -0.19$, $\chi^2(1, 4) = 31.15$,

[4] Congruency (e.g., Incongruent - Congruent) has been used by Lavie et al. (2004). For the sake of consistency, we use this same measure here. We note that this is not the measure (e.g., interference) used in Experiment 1. However, a survey of the literature, and an examination of our results show that: Congruency and interference give roughly parallel results.

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Table 7
Experiment 2: ANOVA on Mean RT, for the Aggregate and Separately for Each of the Observers

O	Effect	*df*	*MSE*	*F*	O	Effect	*df*	*MSE*	*F*
A	L	1	3139	2.85					
	C	1	10194	17.38†					
	L × C	2	13.32	460.28†					
1	L	1	2860160	27.86‡	2	L	1	6329	<1
	C	1	11342388	110.48‡		C	1	7976	<1
	L × C	2	687727	6.67‡		L × C	2	6315	<1
3	L	1	84858	52.28‡	4	L	1	2741955	38.00‡
	C	2	11279319	205.68‡		C	1	10554535	146.30†
	L × C	2	28160	0.51*		L × C	2	458377	6.35*
5	L	1	344547	5.72	6	L	1	224896	1.61
	C	2	4937395	81.31‡		C	1	4630969	33.07‡
	L × C	2	176590	3.91*		L × C	2	521586	4.72*
7	L	1	534656	8.77†	8	L	1	49441	1.19
	C	2	3153939	51.76‡		C	1	172428	1.15*
	L × C	2	168341	4.76‡		L × C	2	47700	1.15

Note. L = load; C = congruency; A = aggregate; O = observers.
$^{*}p < .05$. $^{\dagger}p < .01$. $^{\ddagger}p < .001$.

$p < .0001$]; however, the ratio for incongruent condition was not, [$\beta = -0.01$, $\chi^2(1, 4) = 0.20$, $p = .65$].

We next consider the capacity analysis at the level of the individual observers. The lower portion of Table 8 presents the results of this analysis. Four of the eight observers showed a reliable load × congruency interaction. As a result, at the level of the hazard function we found many individual differences, with only half of the observers showing performance consistent with the predictions of load theory.

We also estimated the effect of congruency (e.g., the hazard ratio for congruent and incongruent trials) under low and high cognitive load conditions, separately for each observer. The estimates for the proportionality due to congruency are plotted in Figure 5. Several points are notable here. First, the sign of most of the estimates is negative, signaling an increase in processing capacity due to congruency. Second, most of the points lie along and beneath the diagonal.[5] This reflects the fact that larger decreases in processing capacity were observed under high than under low cognitive load. Third, the estimates for low cognitive load seem to correlate with those of the high cognitive load. This suggests that the effects of congruency are qualitatively the same at the two levels of load, with differences being quantitative rather than qualitative, as suggested by load theory.

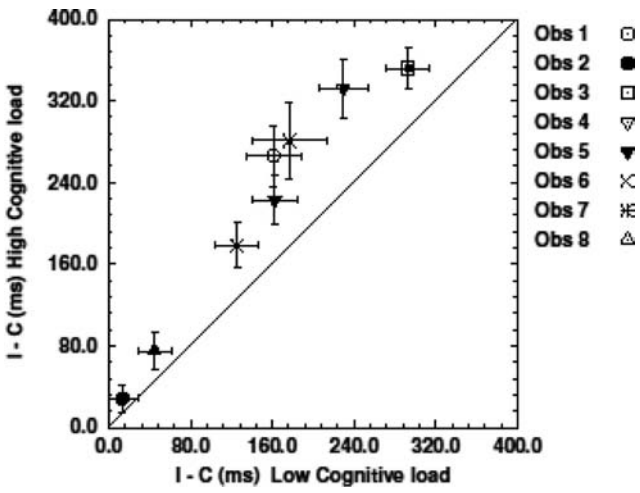


Figure 4. Experiment 2: Differences in mean RTs between congruent and incongruent conditions under low and high cognitive load. Error bars represent the standard error of the mean.

Sensitivity and Response Bias

We next consider the data with respect to Hypotheses 3b and 4b. As mentioned earlier, according to the assumptions of load theory, cognitive load is not predicted to affect either sensitivity or response bias, at either low or high load. Following our earlier analysis in Experiment 1, a "hit" was defined as responding "x" to the target "x" and a "false alarm" was defined as responding "x" to the target "z" (see also Yi et al., 2004).

The pertinent analysis for Hypothesis 3b is presented in Table 10. The aggregate level showed better discriminability under congruent than under incongruent conditions. However, cognitive load did not affect discriminability (see also, Macdonald & Lavie, 2008). Taken together, these results are not fully consistent with Hypothesis 3b. Figure 6a presents the differences in d' (e.g., congruent minus incongruent) in low cognitive load against the same measure in high cognitive load, for each of the observers. As can be seen in the figure, the effect of congruency on d' is substantial. However, observers vary with respect to the magnitude and direction of the difference. Four observers (2, 3, 5, 7), showed larger decreases in d' under low cognitive load than under high cognitive load. Their data points lay beneath the diagonal. Four other observers (1, 4, 6, 8) revealed larger differences in d' under high cognitive load than under low cognitive load with their data laying above the diagonal. These results are inconsistent with Hypothesis 3b.

The analysis of the aggregate response bias data (pertinent to Hypothesis 4b, see Table 10) revealed a reliable effect of cognitive

[5] The diagonal represents equal levels of changes in processing capacity.

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Table 8
Experiment 2: Proportional Hazards Analyses for the Aggregate and Individual Observers

O	Effect	β	*SE*	χ^2	O	Effect	β	*SE*	χ^2
A	L	−0.60	0.07	62.84‡					
	C	0.10	0.03	9.92†					
	L × C	−0.09	0.04	5.92*					
1	L	−0.44	0.10	201.93‡	2	L	0.03	0.13	0.04
	C	0.47	0.09	26.78‡		C	0.08	0.11	0.15
	L × C	0.291	0.132	4.79*		L × C	−0.06	0.16	0.15
3	L	−0.09	0.09	0.95	4	L	−0.41	0.10	17.21‡
	C	1.10	0.09	138.23‡		C	0.69	0.09	55.24‡
	L × C	0.35	0.12	7.83†		L × C	0.19	0.14	1.82
5	L	−0.09	0.11	0.79	6	L	−0.06	0.10	0.33
	C	0.24	0.10	4.58*		C	0.23	0.09	5.87*
	L × C	−0.30	0.09	11.71‡		L × C	0.34	0.142	5.73*
7	L	−0.21	0.08	6.16*	8	L	−0.12	0.08	2.15
	C	0.35	0.08	16.97‡		C	0.06	0.08	0.52
	L × C	0.14	0.12	1.29		L × C	0.12	0.12	1.05

Note. L = load; C = congruency; A = aggregate; O = observers.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

load on response bias. Specifically, observers evidenced a relatively more liberal response criterion when performing under low cognitive load than with high cognitive load. This result is inconsistent with Hypothesis 4b because it implies that cognitive load induces data limitations in the perceived target stimuli. Figure 6b presents the differences in c under high and low cognitive load for the individual observers. The individual data reveal effects of congruency on bias that were not captured by the overall ANOVA. Several patterns are discernible. Two observers (4, 5) evidenced relative liberal shifts in bias both under low and high cognitive load. Two participants (6, 8) showed no reliable differences whatsoever. One observer (3) showed a relative liberal shift in bias under low cognitive load, but a relative conservative shift under high cognitive load. Two observers (1, 2) showed no reliable differences whatsoever. Unlike the aggregate analysis, the individual data revealed extremely variable patterns of changes in response bias. Changes in bias are not predicted by the load theory. In sum, neither the aggregate analysis nor the individual data support Hypothesis 4b.

Table 9
Experiment 2: Estimated Parameters for Changes of Capacity Under Cognitive Load, for the Aggregate and for the Individual Observers

Observer	Load	β	*SE*	χ^2	% Change
A	Low	0.45	0.03	187.32‡	56
	High	0.64	0.03	328.40‡	90
1	Low	0.500	0.088	31.59‡	65
	High	0.715	0.093	58.33‡	100
2	Low	0.075	0.11	0.44	0.78
	High	0.03	0.11	0.065	0.30
3	Low	1.233	0.095	165.126‡	243
	High	1.558	0.106	213.24‡	375
4	Low	0.677	0.094	51.33‡	97
	High	0.930	0.108	73.62‡	154
5	Low	0.511	0.094	29.55†	67
	High	0.732	0.099	54.12‡	108
6	Low	0.246	0.096	6.52*	28
	High	0.57	0.107	28.34‡	77
7	Low	0.343	0.086	15.81‡	41
	High	0.521	0.089	33.64‡	68
8	Low	0.057	0.084	0.46	5
	High	0.172	0.087	3.88*	18

Note. A = aggregate; O = observers.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

Discussion

Experiment 2 provided a test of four hypotheses pertinent to the effect of cognitive load derived from load theory (Lavie et al., 2004). The first was the mean RT pattern observed in previous studies. Three other hypotheses were novel and addressed capacity, sensitivity, and response bias. Table 11 summarizes the results for all four hypotheses, at the level of the aggregate and the level of the individual observers.

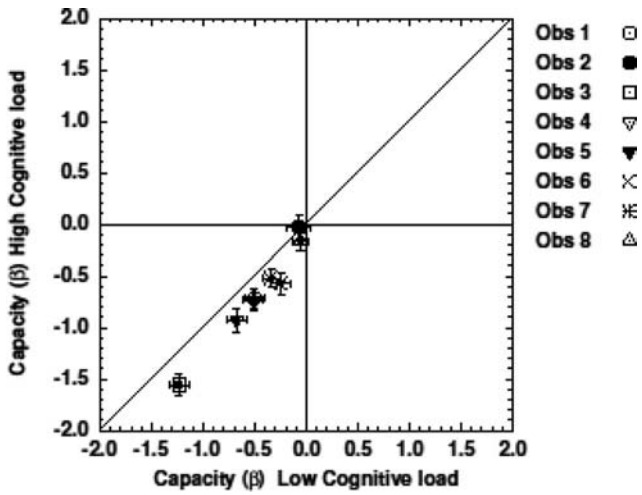


Figure 5. Experiment 2: Capacity coefficients (β) for low- and high-cognitive load.

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Table 10
Experiment 2: Results of the Aggregate Analysis of the Sensitivity and Bias Measures

Measure	Effect	*df*	*MSE*	*F*
d′	L	1,7	0.65	0.08
	C	1,7	0.48	12.58†
	L × C	1,7	0.50	0.16
c	L	1,7	0.018	9.12*
	L × C	1,7	0.01	0.13

Note. *d′* = sensitivity; C = bias; L = load; C = congruency.
* $p < .05$. † $p < .01$. ‡ $p < .001$.

Hypothesis 1b specified the necessary conditions for making the current results relevant to and consistent with previous studies of cognitive load (Lavie et al., 2004). The hypothesis assumed that at the level of the mean RT, the regularities predicted by the load theory would hold. We found the interaction of cognitive load and congruency—better selective attention under low than under high cognitive load—to be reliable at the level of the aggregate. This finding supports Hypothesis 1b. However, in a similar fashion to the perceptual load results, we found substantial variability at the level of the individual observers. Five out of our eight observers did not show a reliable load × congruity interaction. This cannot be attributed to low statistical power or failure to affect cognitive load for these participants, because those factors were carefully controlled. Each observer participated in more than 1,700 experimental trials.

Hypothesis 2b has to do with the effects of cognitive load on processing capacity. According load theory, lower levels of working memory load are associated with better control over irrelevant distractors, leading to critical differences in capacity. Hypothesis 2b thus took the form of a set of orderings on the hazard functions of the RT distribution. At the level of the aggregate, all of the predictions of Hypothesis 2b were confirmed. The congruency effects on capacity were observable but less pronounced at low levels of cognitive load than at high levels of cognitive load, $h_{H,C}(t) > h_{H,I}(t)$ and $h_{L,C}(t) > h_{L,I}(t)$. As predicted by Hypothesis 2b, the magnitude of these differences was ordered, such that: $\frac{hH,C(t)}{hH,I(t)} > \frac{hL,C(t)}{hL,I(t)}$. However, we found substantial variability at the level of the individual observers. This ordering was reliable for only four out of eight participants. In addition, we found only partial support to the predicted ordering on the hazard functions across levels of load, holding congruency fixed, [i.e., $h_{H,C}(t) < h_{L,C}(t)$ and $h_{H,I}(t) < h_{L,I}(t)$] with only the former holding.

Hypotheses 3b and 4b assumed that cognitive load does not diminish the informational value of the stimulus. This prediction was tested by estimating the sensitivity and bias values under all experimental conditions. Cognitive load did not affect sensitivity, but congruency did with no interaction of load and congruency. In addition, cognitive load affected response bias both at the aggregate level and at the individual level. This pair of results is not consistent with the lion part of neither Hypothesis 3b nor Hypothesis 4b.

Taken together, the hypotheses pertinent to cognitive load both at the aggregate and at the individual level (see Table 11) cast serious doubts on the consistency and generality of load theory central assumption.

General Discussion

The present study provided a close look at a prominent, contemporary theory of selective attention: load theory (Lavie et al., 2004). We used the assumptions of load theory to derive eight hypotheses, six of which were novel to the present effort. Hypotheses 1a–4a referred to the role of perceptual load in selective attention. Hypotheses 1b–4b referred to the role of cognitive load (see Table 1). We relied on the logic of converging operations (Garner, Hake, & Eriksen, 1956) across a set of dependent measures (mean RT, hazard functions, sensitivity, and response bias) at two levels of analysis (aggregate and individual observer). This approach has a number of precedents generally (e.g., Estes, 1956; Maddox, 1999) and in the study of selective attention (e.g., Yee & Hunt, 1991).

Let us briefly recap the hypotheses and the relevant findings. Hypothesis 1a and 1b concerned effects of perceptual and cogni-

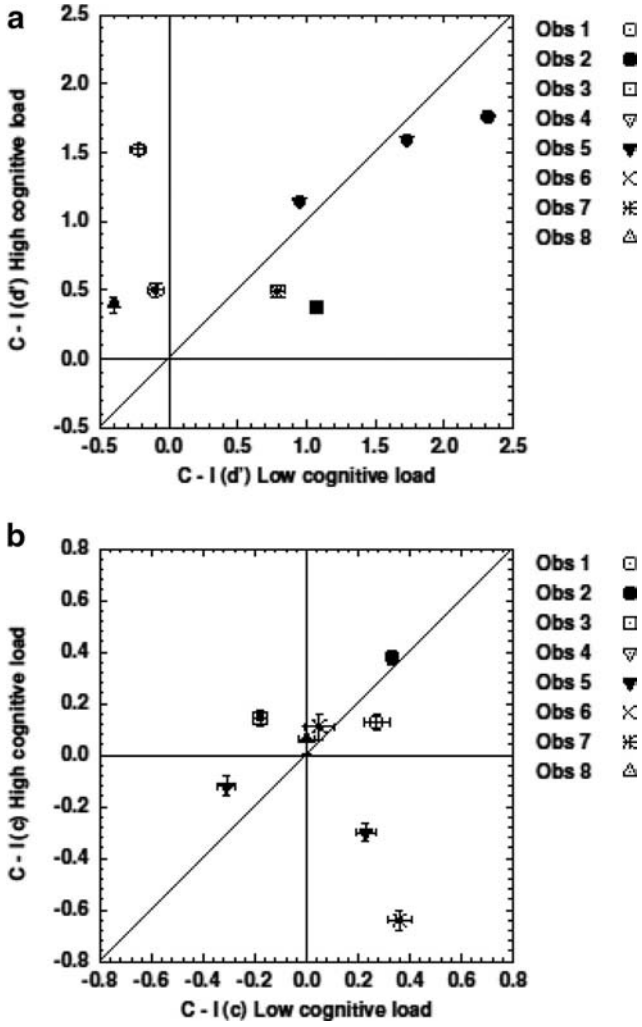


Figure 6. Experiment 2: differences in sensitivity (*d′*; Panel a) and in response bias (Panel b) between congruent and incongruent conditions for individual observers under low and high levels of cognitive load. Note: error bars indicate 95% confidence intervals (see for Gourevitch and Galanter, 1967).

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Table 11
Experiment 2: Summary of the Outcomes With Respect to Each of the Four Hypotheses

	Hypothesis			
Observer	1(a)	2(a)	3(a)	4(a)
Aggregate	∨	∨	*	*
1	∨	∨	*	*
2	*	*	*	*
3	*	∨	*	*
4	∨	*	*	*
5	∨	∨	*	*
6	∨	∨	*	∨
7	∨	*	*	*
8	*	*	*	∨

∨ prediction holds. * prediction fails.

tive load on mean RT. The former implied better selective attention under low rather than under high perceptual load. The latter predicted a reversed pattern—improved selectivity under low relative to high cognitive load. These opposite effects are considered to be the empirical signatures of load theory (Lavie et al., 2004). The results of the current experiments supported these hypotheses at the level of the aggregate. However, the evidence was much more ambiguous at the level of the individual observers.

Hypotheses 2a and 2b examined assumptions with respect to processing capacity and processing efficiency, allegedly mental variables pertinent to perceptual and cognitive load effects. Framed in terms of orderings on the RT hazard functions, these hypotheses allowed direct assessments of those psychological constructs, as well as fine-grained analyzes on the entire RT distribution (Townsend & Ashby, 1978; Wenger & Gibson, 2004). The hypotheses were generally supported at level of the aggregate. Critically, however, we found considerable variations at the level of the individual observers. Hypotheses 3a, 3b, 4a, and 4b concerned the extent to which sensitivity and response bias would be influenced by perceptual and cognitive load. Lavie and her colleagues have forcefully argued that perceptual (and cognitive) demands induce resource limitations rather than data limitations. By their argument, load exclusively affects capacity while leaving the target's informational (e.g., data) value intact (Lavie & de Fockert, 2003; Lavie, 2005). These hypotheses were expressed in terms of equalities on signal detection parameters d' and c. Contrary to the assumptions of load theory, these hypotheses were not supported at either the level of the aggregate or the level of the individual observer.

Overall, the evidence in favor of load theory was mixed. Tables 6 and 11 give a summary of the accumulated evidence from Experiment 1 and Experiment 2. As can be seen in these summary tables, support for load theory was obtained at the level of the aggregate for the mean RT (speed of processing) and the hazard function (capacity). The evidence against the load theory came at the level of the individual observers for these same two dependent measures. These violations of the assumptions of load theory suggested that data limitations (Norman & Bobrow, 1975) were operative. In Experiment 1, we found reliable changes in sensitivity and response bias due to perceptual load. In Experiment 2, we found changes in response bias due to cognitive load and congruency. To account for these outcomes, one must assume some degradation in the quality of the stimulus information (e.g., data) in addition to or as a result of changes in processing capacity. Our conclusions get further support from a recent study by Yi et al. (2004). Although the results of this work were not reported in terms of response sensitivity and bias, the authors did (graphically) report the hit and false alarm rates (Figure 3, p. 994), making it possible to estimate d' and c. Doing this revealed that the manipulations of load produced decrements in sensitivity and shifts in response bias, generally consistent with those obtained in the present study.

Implications for Theories of Load and Selective Attention

Two prominent classes of theories account for the effects of perceptual and cognitive loads on selective attention. One is load theory (Lavie, 1995; Lavie et al., 2004), which relies on the notion of resources or processing capacity. Another class of theories, based on properties of neural circuits, and on the idea of lateral inhibition in particular, has been proposed by LaBerge (LaBerge, 2002; LaBerge, Brown, Carter, Bash, & Hartley, 1991). According to LaBerge, failures of selective attention (e.g., as gouged by distractor processing) are driven by lateral inhibition that spreads outward from target sites to distractor site. To account for load effects, LaBerge assumes that inhibition weakens as attentional activity at the target site decreases (e.g., due to increase in perceptual load or decrease in cognitive load), and consequently leads to enhanced distractor processing. Both load theory and LaBerge's inhibition theory agree that target and distractor get differential amounts of processing under various levels of load. However, the theories disagree on the mechanisms that produce those processing levels. Moreover, load theory assumes the existence of two separate and independent mechanisms, one for the processing perceptual load, the other for the processing of cognitive load, whereas the inhibition theory assumes a single mechanism for both.

Recently, a novel explanation has been suggested by Tsal and Benoni (2010) to account for perceptual load effects. Their account is based on the idea of dilution (Kahneman & Chajczyk, 1983), and it diverges radically from the former theories in various important respects. According to Tsal and Benoni, features of neutral letters (e.g., high perceptual load) compete with those of the incongruent distractor at an early visual stage, thus degrading the quality of its visual representation. This reduction in discriminablity hampers its lexical processing, and consequently leads to reduction, or elimination of the interference effect. To test for their hypothesis, Tsal and Benoni have used three types of displays: low perceptual load–low dilution, high perceptual load–high dilution, and low perceptual load–high dilution. The first two displays are the traditional low and high perceptual load displays, used in previous studies. The third display consisted of a target appearing in a distinctive color, and embedded among several neutral letters. The critical result, one that confirmed the dilution hypothesis, was the elimination or reduction of interference in the low perceptual load–high dilution condition.

Taken together, the results of the present study can not be completely accommodated by any of the proposed theories discussed above: load theory (Lavie et al., 2004), lateral inhibition theory (LaBerge, 2002; LaBerge et al., 1991), or the dilution

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theory (Tsal & Benoni, 2010). Indeed, we found some evidence in favor of load theory, and the lateral inhibition hypothesis regarding the effects of load on target and distractor processing. However, when tested at the individual level, some observers did not exhibit those effects. Our capacity measures supported the load theory hypothesis regarding processing capacity. However, again, individual differences suggested that changes in processing capacity might not be the only mechanisms at work. This conclusion was further supported by the finding that signal detection measure d' for target classification was greater under low perceptual load than under high perceptual load. This entails that perceptual load does not exclusively influence the level of processing capacity, but also affects the discriminability of the target. This outcome is not predicted by load theory (Lavie, 1995), but is fully consistent with Tsal and Benoni's (2010) dilution account. At their current stage of development neither load theory nor the dilution theory can account for load-related changes in response bias, found in the current study. LaBerge (2002) inhibition theory can accommodate those results by assuming that changes in the strength of inhibition causes shifts in response strategy.

In our view, load affects both the informational value (e.g., data limitations) and the allocation of processing capacity (e.g., resource limitations) of the target. We argue that the distinction made by load theory between data limitations and resource limitations (Lavie & de Fockert, 2003) is too strict, and reflects a misinterpretation of the original theory (Norman & Bobrow, 1975). In their seminal paper Norman and Bobrow asserted that data and resources can be affected simultaneously to a certain point (for a similar argument see also, Tsal & Benoni, 2010). It seems highly probable from our data that both resources (e.g., processing capacity) and data (e.g., discriminablity and response bias) are influenced by manipulations of load. Interestingly, the capacity measures in our data were not correlated with any of the other SDT measures, suggesting a functional independence of resource and data. Another open theoretical question is whether perceptual and cognitive load necessitate two separate explanatory mechanisms. Our results did not provide a clear answer to this question, but we endorse the view that it would be more parsimonious to assume a single mechanism as assumed for example by LaBerge (2002).

Aggregate Versus Individual Results

A critical question raised by the results of the current study concerns the discrepancies between the results at the level of the aggregate—generally supportive of load theory—and the results at the level of the individual observers. This, in and of itself, is not a novel finding. A number of studies (e.g., Estes, 1956; Maddox, 1999; Townsend & Fific, 2004; Feldman, Barret, Tugade & Engle, 2004; Siegler, 1987; Yee & Hunt, 1991) have documented such discrepancies. Why might they be obtained? One possibility is that individual observers may be employing different response strategies, and doing so consistently. Another possibility is that individual observers may vary in terms of the strategy that they employ at any given time during the experiment. In either case, averaging across these variations may produce data consistent with the general patterns predicted by load theory, even though that specific pattern is not produced (or not produced consistently) by any one observer (see Estes, 1956; Maddox, 1999; Townsend & Fific, 2004; Siegler, 1987, for detailed discussions).

Either of these possibilities might suggest that load theory's conception of performance under varying levels of demand is incorrect. We would, however, suggest an alternative interpretation: Load theory's ability to describe performance in the aggregate, but not consistently at the level of individual observers, is a function of the theory's "limited vocabulary": that is, the constructs of load theory capture a generality without having the flexibility to predict variations in performance across or within observers. From this perspective, the failure to account for the range of differences across observers can be addressed by extending the theoretical specificity, and thus the range, of empirical possibilities.

One possible way to do this would be to consider a modeling approach capable of characterizing variations in the fundamental characteristics of information processing. For example, work with meta-theories such as general recognition theory (Ashby & Townsend, 1986) and systems factorial theory (Townsend & Nozawa, 1995) has demonstrated the potential for predicting variations in experimental outcomes on the basis of individual differences in performance (e.g., Copeland & Wenger, 2006; Ingvalson & Wenger, 2005; Wenger & Townsend, 2001, 2006). An approach capable of capturing these general distinctions, that can simultaneously address predictions for response frequencies and latencies (see Townsend & Wenger, 2004a,b), and that can systematically model the possible effects of varying types of load at different stages of processing would allow for the exploration of such possibilities. Thus, we suggest that the discrepancies between the aggregate and individual observer data suggest a need to expand the theoretical range of load theory, rather than suggesting that the theory itself is critically flawed.

Conclusions

The work presented here suggests critical problems with load theory, as it is currently framed. Specifically, the evidence suggesting data limitations and the presence of both quantitative and qualitative differences across observers calls into question the ability of load theory in its present form to account for the experimental results. We strongly suggest that this implies a need to expand the "vocabulary" of load theory, rather than a need to discard it, given the relative success of the approach at capturing aggregate regularities.

To do this, we would suggest, will require that the form of the theory be expanded to allow for predictions for the range of dependent variables used in the present study. The present work illustrates the implications for multiple converging measures implicit in load theory. The next step is to adopt a vocabulary that allows that potential to be made explicit.

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